Manankumar Thakkar

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SUMMARY

Software Engineer with 4+ years of experience designing scalable, distributed systems in Java and Spring Boot and building generative AI agents and LLM tools. Builds multi-threaded microservices, Python data pipelines, and LLM applications with typed outputs, ground-truth evaluation, and cost controls. Proven in production: cut API latency **60%** and prevented a **6x data collision**.

TECHNICAL SKILLS

Languages: Java, Python, SQL, TypeScript, JavaScript
Backend: Spring Boot, Spring MVC, Hibernate/JPA, Microservices, Distributed systems, REST APIs, Concurrency, Kafka, Redis, Flask, Node.js, Liquibase
Frontend: React, Next.js, Tailwind CSS, Three.js
AI / LLM: Generative AI agents, LLM evaluation, Structured outputs (Zod), RAG, MCP / tool use, LangChain, Ollama
Cloud & Data: Kubernetes, Docker, GCP (BigQuery, GCS), Azure SQL, PostgreSQL, MySQL, AWS, Vercel, Terraform
SDLC & Ops: CI/CD, GitHub Actions, Git, JUnit, Vitest, Playwright, Postman, Grafana, Prometheus, Splunk, SonarQube, Snyk

EXPERIENCE

Walmart Global Tech | Software Engineer III **02/2025 – Present**

- Expanded advertiser booking flexibility **40%** by architecting in-store ad scheduling for Walmart Vision Centers in **Java/Spring Boot**, with per-location ad-rotation rules and booking by seconds of screen time instead of percentages.
- Cut API latency **60%** and **10,000+ database calls**/day by fixing an N+1 bottleneck with concurrent caching in **microservices**.
- Prevented a **6x data collision** in a multi-tenant expansion into new international markets by catching a flawed data-model key for ad-scheduling settings across 7+ partner teams and driving a durable country-level key over a regional patch.
- Automated readiness reporting for in-store ad screens across **4,000+ stores** twice daily with a **Python** pipeline (Azure SQL, Redis, BigQuery, Tableau) applying 13 business rules; matched the manual report and caught **3 defects**.
- Built a **generative AI operations assistant** (Python, Flask, LangChain) on Walmart's internal LLM Gateway, giving engineers chat access to operational context through a REST API with session-scoped memory and response caching.

Walmart Global Tech | Software Engineer **07/2024 – 01/2025**

- Extended operating hours across **4,000+ stores** to Deli, Bakery, and Service Deli departments with **Java/Spring Boot** services, versioned Azure SQL migrations via Liquibase, and batch lookups replacing per-store queries.
- Enabled targeted in-store advertising for new store formats by grouping each location's digital screens across three **REST APIs**.

George Mason University | AI Research Assistant **07/2023 – 07/2024**

- Developed research tooling powered by **LLMs** (ChatGPT) that automated and streamlined manual research workflows.
- Provisioned and managed research infrastructure as code with **Terraform** for reproducible research environments.

Nokia | Test Automation Engineer (Co-op) **09/2022 – 06/2023**

- Reduced manual testing effort **50%** by building **Python test automation** suites for high-throughput systems.
- Cut defect density **20%** and improved code quality metrics **35%** by integrating automated tests into **Jenkins CI/CD** pipelines.

Let It Wag | Full Stack Developer **06/2019 – 05/2020**

- Improved backend efficiency **20%** while building **Node.js** services and REST APIs that handled real-time data for **13K+ users**.
- Delivered **90%** of projects on time while building a responsive **React** UI supporting 100+ user journeys.

AI PROJECTS

Archwoven | *Next.js, TypeScript, Anthropic/Ollama, tree-sitter* | Live | GitHub

- Shipped a live **AI architecture tool** that turns system descriptions or GitHub repos into 4 editable diagram views, passing LLM output through Zod schemas, graph checks, and bounded repair before rendering.
- Matched hand-checked ground truth on Spring PetClinic (**19/19 nodes**, **10/10 edges**) via tree-sitter extraction; reserved worst-case cost before every paid LLM call and shipped **1,000+ tests**.

MCP Security Observatory | *Python, tree-sitter, GitHub Actions* | Live | GitHub

- Engineered a static scanner that audits **MCP servers** without executing them, covering **1,600+ servers** from a 26,000+ repository corpus with 5 tree-sitter taint rules (shell exec, path traversal, tool-description injection, hidden Unicode, scope).
- Enforced responsible disclosure in code, withholding **800+ high/critical findings** until maintainers are notified; shipped a nightly pipeline, public dashboard, SARIF output, and **782 tests** with ruff, mypy, and CodeQL in CI.

EDUCATION & CERTIFICATIONS

George Mason University, Fairfax, VA – *Master of Science in Computer Science* *08/2021 – 05/2023*

- Walmart Applied Agentic AI: Foundation to Production (completed with capstone)